Discrete Probability Theory (IN0018)
School of Computation, Information and Technology
Technical University of Munich

Tutorial 07

Conditional Expected Value, Convolution, Inverse Transform Sampling

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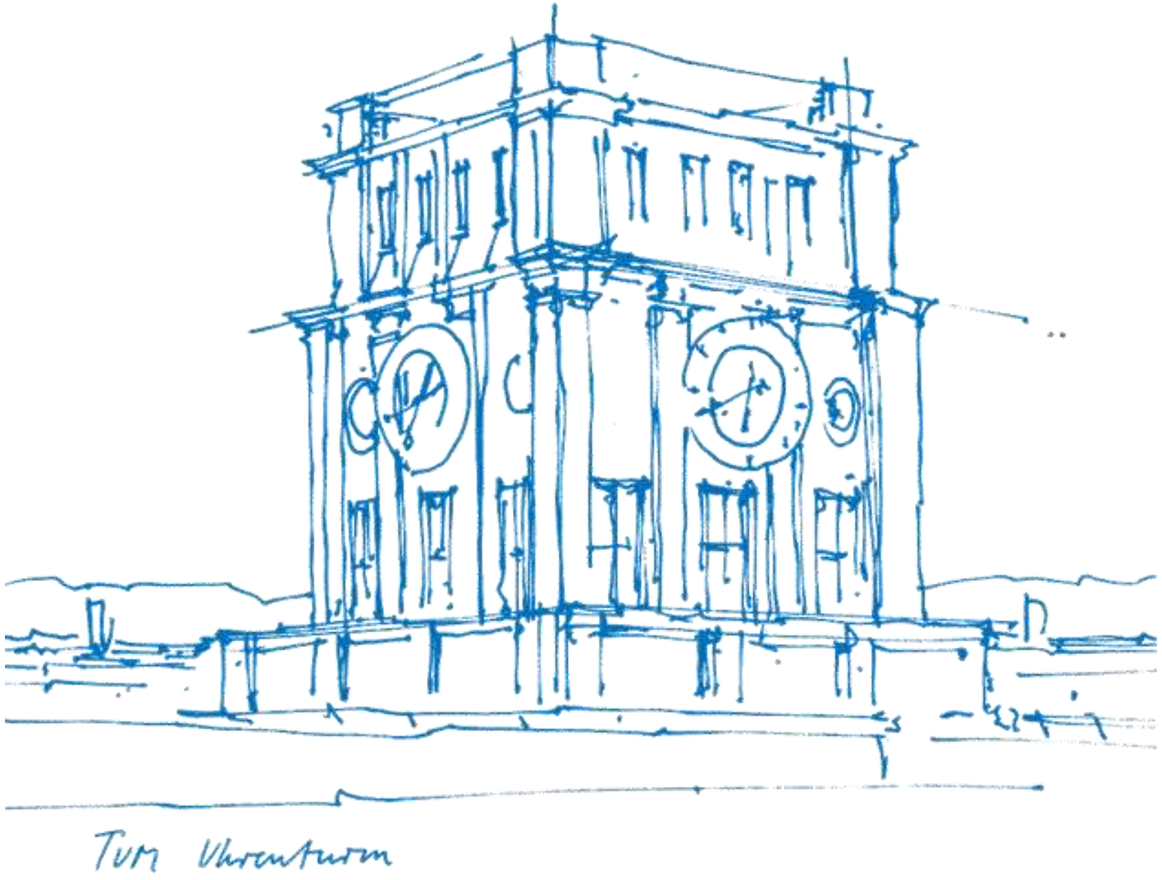

About Tutorial

- **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
  - 14:15-15:45 Tue, 00.08.053 (B42)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code ☞

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➢ PART I. Recap of the Lecture

➢ PART II. Exercises from the Exercise Sheet

➢ PART III. Additional Exercises (with solution)

I. Recap

- **Conditional Expected Value**
  - Definition
- **Convolution**
  - Definition
- **Inverse Transform Sampling**
  - Definition

II. Exercise Sheet

Exercise 1. Let $(\Omega, \mathcal{A}, P)$ be a probability space, $X, Y : \Omega \to \mathbb{R}$ two $\mathbb{R}$-valued RVs with joint density

$$p_{X,Y}(x, y) = \exp(-y)\chi_A(x, y) \quad \text{for } A = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq y\} \,.$$

We can also write this equivalently as

$$p_{X,Y}(x, y) = \begin{cases} \exp(-y) & \text{if } x, y \in \mathbb{R} \text{ and } 0 \leq x \leq y \,, \\ 0 & \text{otherwise} \,. \end{cases}$$

(i) Verify that $p_{X,Y}(x, y)$ is a valid pdf.

(ii) Compute the conditional expectation $\mathbb{E}[X \mid Y]$.

(iii) Compute the conditional expectation $\mathbb{E}[Y \mid X]$.

Exercise 2. Let $(\Omega, \mathcal{A}, P)$ be a probability space, $X, Y : \Omega \to \mathbb{R}$ two independent RVs with densities $p_X, p_Y : \mathbb{R} \to \mathbb{R}_{\geq 0}$.

(i) Let $Z := X + Y$. Then compute the pdf of Z and show that it is given by

$$p_Z(z) = \int_{\mathbb{R}} p_X(z-y)\, p_Y(y)\, dy\,.$$

Remark: Generally, for two functions $f, g : \mathbb{R} \to \mathbb{R}$, we call $(f \star g)(t) := \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,d\tau$ the *convolution* of f and g. (Note that this is symmetric.) Hence, $p_Z(z)$, the density of the sum of X and Y, is given by the *convolution* of the densities $p_X(x)$ and $p_Y(y)$ of X and Y. Accordingly, one sometimes also calls Z the convolution of X and Y (even though it's just the sum of X and Y and we didn't really need an additional name).
Note: This also holds more generally for independent $\mathbb{R}^n$-valued random variables.
[Hint: Consider the set $A_z := \{(x,y) \in \mathbb{R}^n \mid x + y \leq z\}$ and compute $F_Z(z) = P(Z|lez)$ using the independence of X, Y. Finally, remember that by definition $F_Z(z) = \int_{-\infty}^{z} p_Z(z)\,dz$.]

(ii) Let $X, Y \sim \mathrm{Exp}(\lambda)$ be independent with $\lambda > 0$. Calculate the density of $X + Y$.

(iii) Let $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$, $Y \sim \mathcal{N}(\mu_y, \sigma_y^2)$ be independent with $\mu_x, \mu_y \in \mathbb{R}$ and $\sigma_x, \sigma_y > 0$. Show that

$$X + Y \sim \mathcal{N}(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)\,.$$

This is a fairly long and tedious computation, which would be too lengthy and error prone for the exam, but is a good exercise to get a feeling for how computations with Gaussians work. Start from the convolution on one side and the final result—the density of $\mathcal{N}(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$—and try to "match them from both directions."

II. Exercise Sheet

Exercise 3. Let $(\Omega, \mathcal{A}, P)$ be a probability space and X a random variable on Ω.

(i) Let $X : \Omega \to \mathbb{N}$ be a discrete random variable. Show that

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} P(X \geq n)$$

and describe in words in a single sentence what that means.

(ii) Assume now that $X : \Omega \to \mathbb{N}$ is geometrically distributed with parameter $p \in (0, 1)$, $X \sim \mathrm{Geo}(p)$. Use (i) to calculate the expectation $\mathbb{E}[X]$.
[Hint: Recall the partial sum formula of the geometric series.]

(iii) If $X : \Omega \to \mathbb{R}_{\geq 0}$ is a non-negative, continuous RVRV with cdf F_X, possessing a density function $p : \mathbb{R} \to \mathbb{R}_{\geq 0}$, then it holds that

$$\mathbb{E}[X] = \int_0^{\infty} 1 - F_X(x)\, dx\,.$$

Using this, calculate the expectation of a $X \sim \mathrm{Exp}(\lambda)$, where $\lambda > 0$.

II. Exercise Sheet

Exercise 4 (application + coding). Consider the following (oversimplified) setting: We are at the conveyor-belt of a fishing trawler and should distinguish between different types of fish. We want to separate trouts ("Forelle"), which is the "on average smaller fish") from salmon ("Lachs"), the "on average larger fish." We label trouts as $Y = 0$ and salmon as $Y = 1$. We distinguish them by size, which we denote by X. We also call X a "feature" here. We assume that we assess size with some value in $[0, 1]$, where 0 is smallest and 1 is largest. Mathematically, this can be modeled in the following way. Let X and Y be $[0, 1]$-valued and $\{0, 1\}$-valued random variables on the same probability space, respectively. We assume that there are equally many trouts and salmons in the water, which we model by the prior probabilities

$$P(Y = 0) = P(Y = 1) = \frac{1}{2}\,.$$

Moreover, from experience we know how the sizes of trouts and salmons are distributed. We assume this to be given by the likelihood densities

$$p(x \mid Y = 0) = \begin{cases} 2 - 2x & \text{if } x \in [0, 1]\,, \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad p(x \mid Y = 1) = \begin{cases} 2x & \text{if } x \in [0, 1]\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

We want to use inverse transform sampling to sample X for each class $Y \in \{0, 1\}$. We proceed step by step.

Work out the required inverse transformation formulas "on paper".

III. Additional Exersices

Exercise 5. Use inverse transform sampling to sample from a 2-dimensional Gaussian with

$$\mu = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

More precisely, derive the transformation to sample from a one dimensional Gaussian $\mathcal{N}(0, 1)$.

For know U_1 and U_2, to find $X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$ such that $X \sim N\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\right)$.

Let $U_1, U_2 \sim \text{Uniform}(0,1)$, define $Z_1 = F_Z^{-1}(U_1)$, $Z_2 = F_Z^{-1}(U_2)$ where $Z_1, Z_2 \sim N(0,1)$ and $Z_1 \perp\!\!\!\perp Z_2$,

then $Z = \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, I\right)$.

To find matrix A such that $AA^T = \Sigma$, then we can define $X = \mu + AZ$ and get $X \sim N(\mu, \Sigma)$,

since $E(X) = E(\mu + AZ) = \mu$ and $\text{cov}(X) = \text{cov}(\mu + AZ) = A\,\text{cov}(Z)A^T = AA^T = \Sigma$.

With Cholesky decomposition, solve that $A = \begin{pmatrix} \sqrt{2} & 0 \\ \frac{1}{\sqrt{2}} & \sqrt{\frac{3}{2}} \end{pmatrix}$, so $X = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} \sqrt{2} & 0 \\ \frac{1}{\sqrt{2}} & \sqrt{\frac{3}{2}} \end{pmatrix} \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$,

where $X_1 = 1 + \sqrt{2} Z_1$, $X_2 = 2 + \frac{1}{\sqrt{2}} Z_1 + \sqrt{\frac{3}{2}} Z_2$.

Materials